NEXT GEN SOFTWARE HUB

CYBER SECURITY INTERNSHIP

WEEK 05 TASK 05 REPORT

Vulnerability Scanning

Using Nessus and OpenVAS to Identify and Assess Vulnerabilities

TOPICS

- Vulnerability Scanning
- Risk Assessment

TOOLS

- Nessus
- OpenVAS

RESOURCES

- Nessus Essentials Free
- OpenVAS Guide

TASK

Scan a vulnerable VM and prepare a basic vulnerability report.

Phase 2: Scanning & Penetration Testing (Week 5–10)

PLANS :

Week 5: Vulnerability Scanning



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2026

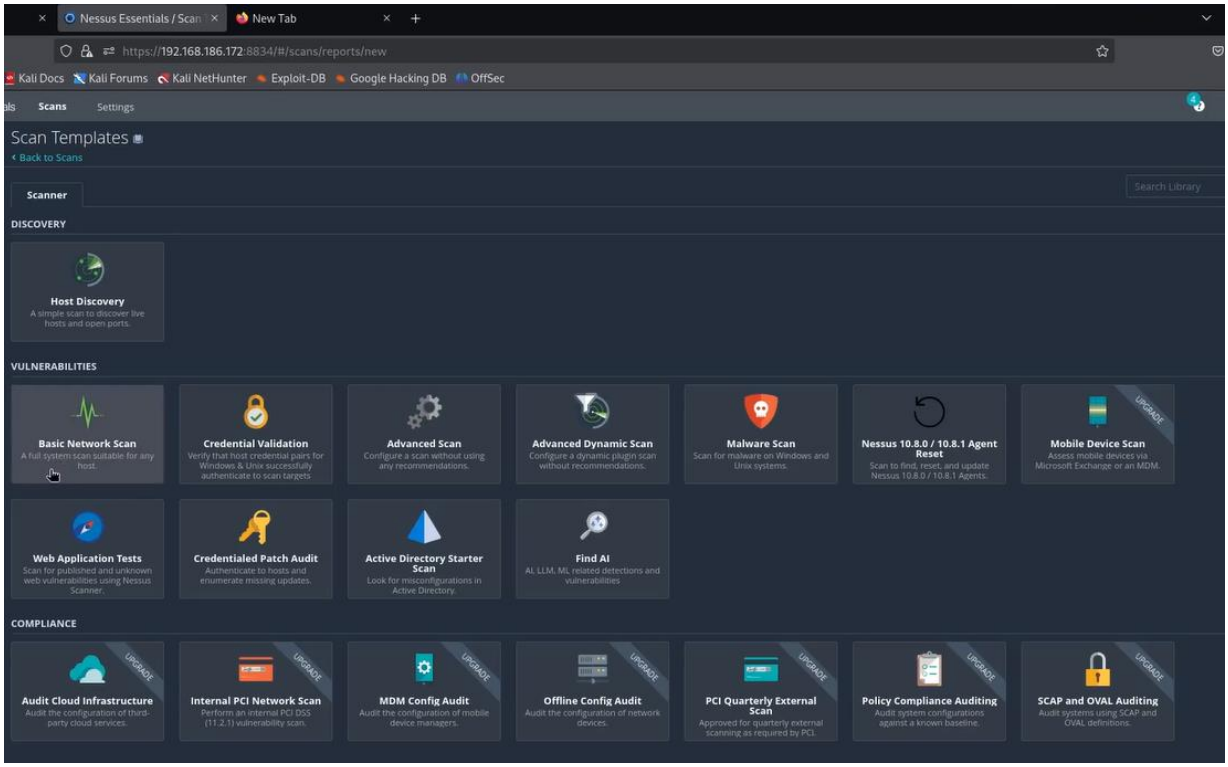
Introduction and Objective

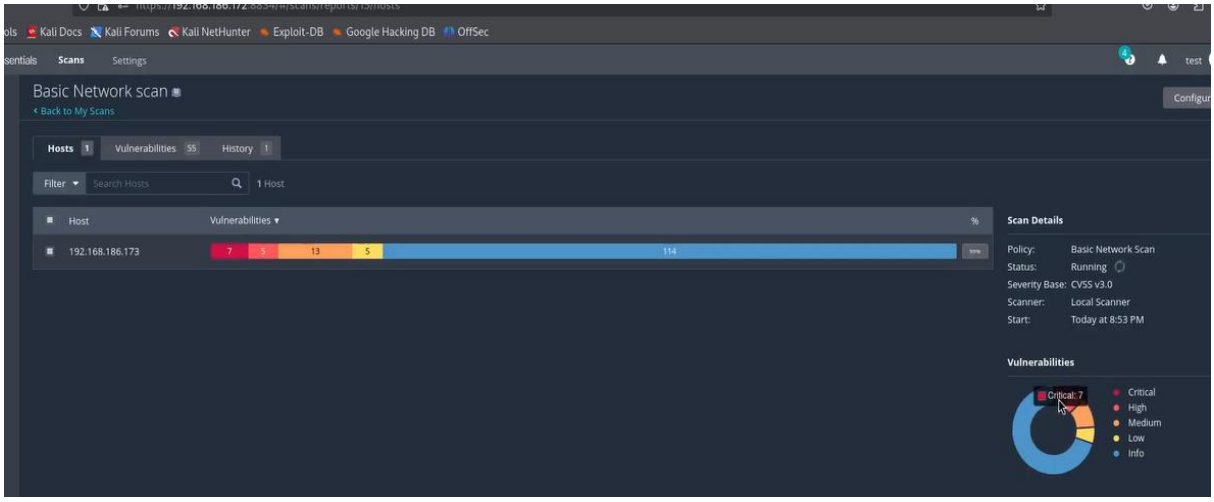
The objective of this task was to perform a comprehensive vulnerability assessment against a target virtual machine using an industry standard scanning tool. While previous tasks focused on discovering open ports and identifying services, vulnerability scanning takes this process a step further by evaluating those services against a massive database of known security flaws. For this assessment, we utilized Nessus Essentials to scan the Metasploitable target machine located at the IP address 192.168.186.173. The goal was to identify critical weaknesses, understand how they can be exploited, and formulate a professional report on how to secure the system.

Methodology and Execution

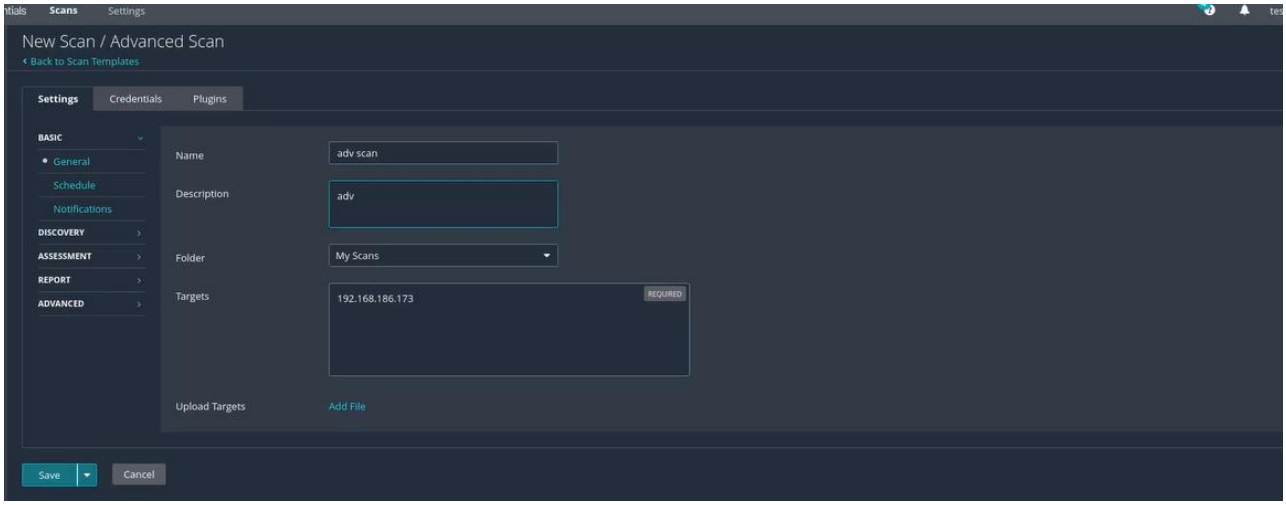
Our assessment was divided into two distinct phases to demonstrate how scanning depth changes with different configurations.

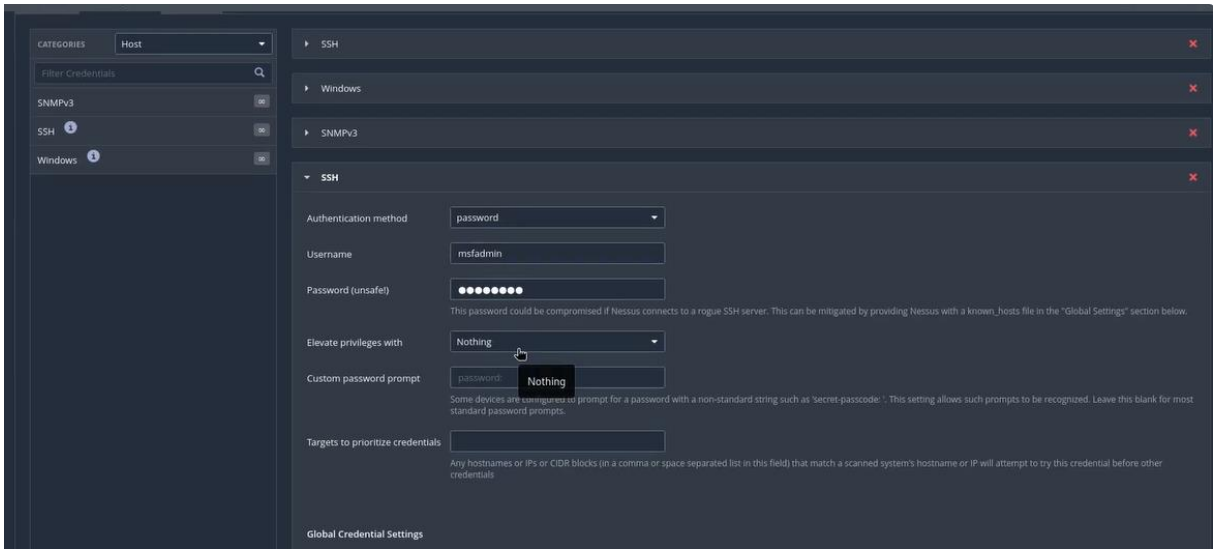
The first phase involved a standard unauthenticated Basic Network Scan. This mimics the perspective of an external attacker who has no login credentials.





The second phase involved an Advanced Network Scan where we provided valid SSH credentials for the target system. By supplying the username and password, Nessus was able to log directly into the machine and inspect the internal operating system and installed packages, which reveals vulnerabilities that are invisible from the outside.

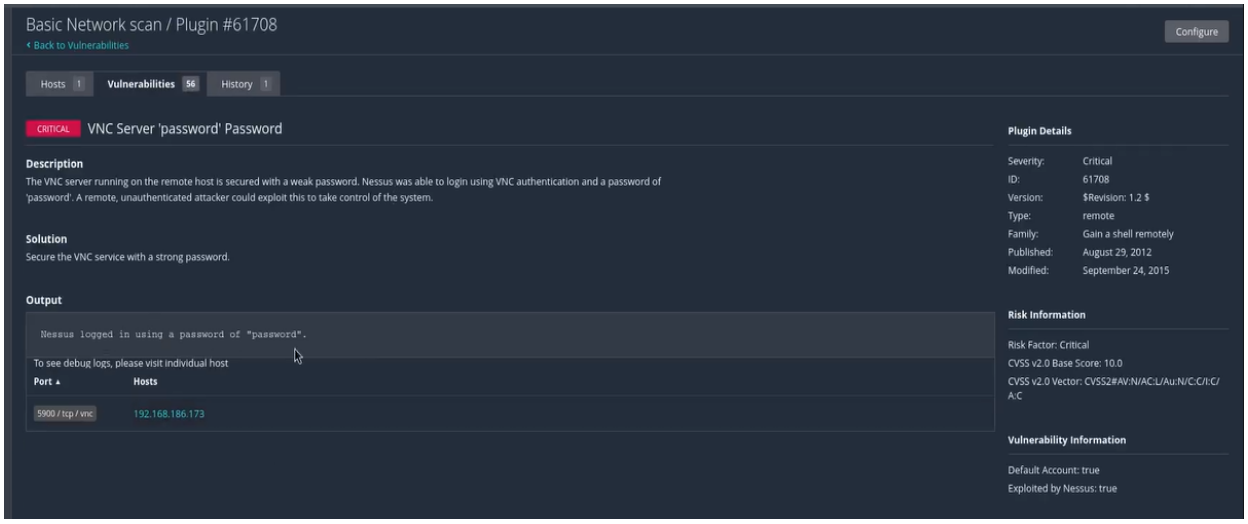




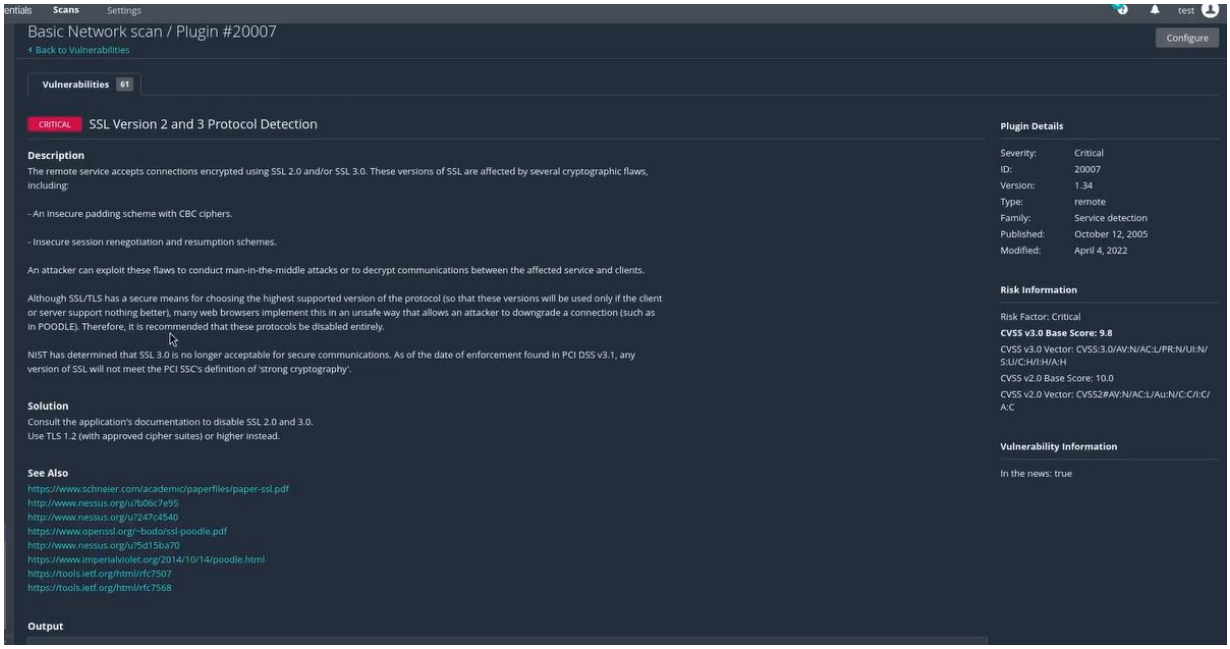
Findings from the Unauthenticated Scan

The initial basic scan was highly successful and returned seventy vulnerabilities. The most severe findings immediately highlighted critical flaws that would allow an attacker to take complete control of the system.

The highest risk finding was a VNC Server password vulnerability. Nessus detected that the VNC service on port 5900 was protected by the default password of "password". Because VNC is a remote desktop tool, this means any attacker on the network could easily guess the password and gain graphical remote control of the machine. This was assigned a perfect CVSS score of 10.0.



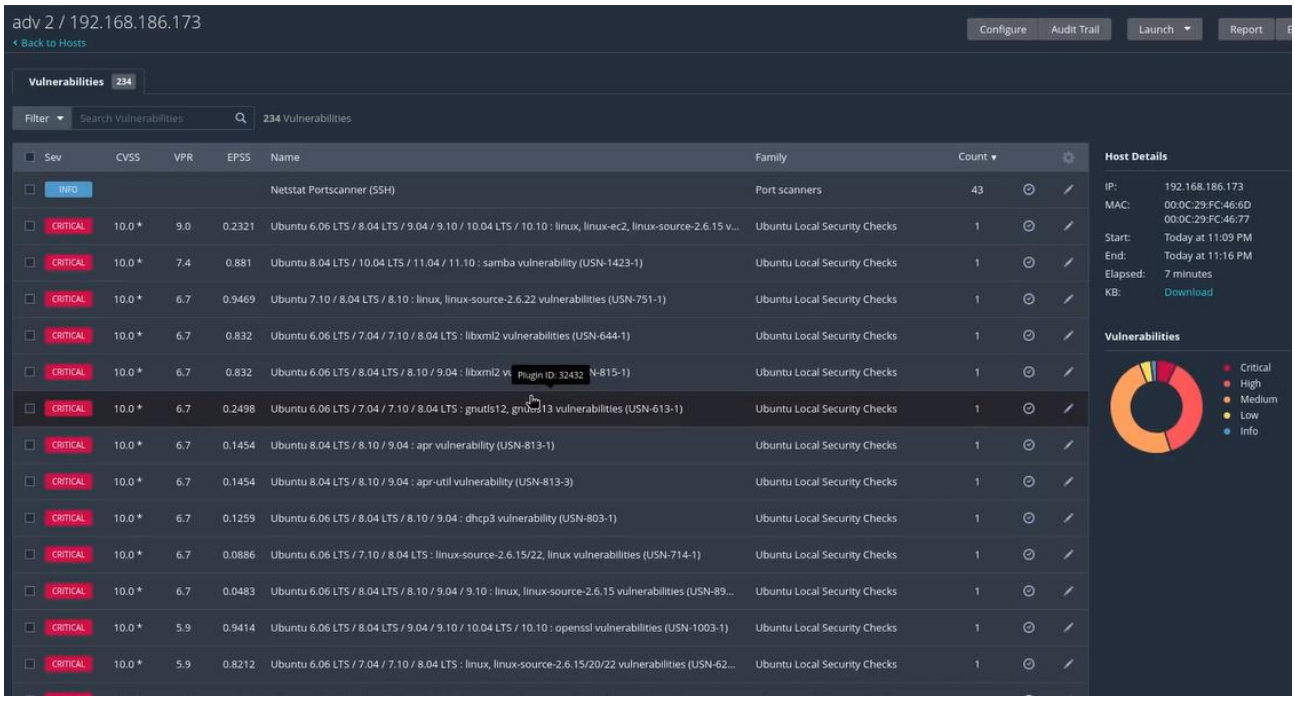
Another critical finding was the detection of a Bind Shell Backdoor. Nessus discovered that a backdoor service was actively listening on the target machine, allowing immediate remote command execution. Additionally, the scanner flagged that the server still supported outdated SSL Version 2 and 3 protocols, which are highly insecure and vulnerable to man in the middle attacks.



Findings from the Authenticated Scan

The second scan, which utilized the SSH credentials, was significantly more revealing. Because Nessus was authenticated, it was able to perform deep local security checks on the Ubuntu operating system. The number of discovered vulnerabilities skyrocketed to two hundred and thirty four.

The results were dominated by dozens of critical severity findings related to the Linux kernel, Ubuntu core packages, Samba, and OpenSSL. These findings demonstrate that the operating system itself is severely outdated and missing years of critical security patches.



Remediation Recommendations

Based on the severity of the findings, immediate action is required to secure this machine. First, the VNC service must either be completely disabled if it is not needed, or immediately updated with a complex, non default password. Second, a full operating system upgrade is mandatory. Applying the latest Ubuntu security patches will resolve the vast majority of the two hundred and thirty four findings discovered during the authenticated scan. Finally, insecure protocols like SSL version 2 and 3 must be explicitly disabled in the server configurations.

Conclusion

This vulnerability assessment successfully demonstrated the immense value of using dedicated scanning tools. We were able to identify severe backdoors, weak passwords, and massive amounts of unpatched software. Furthermore, comparing the unauthenticated and authenticated scans proved a vital cybersecurity concept. An external scan only shows what is exposed to the network, while an authenticated scan reveals the true, full depth of the internal security debt. Regular vulnerability scanning combined with strict patch management is absolutely essential to prevent these types of critical exposures.